

# CLASSIFYING DECELERATION MEASUREMENT SEQUENCES

---

Team **HalloLeute**

Nikita Biswas · Sebastian Garcia · Chaeran Yoo · Daniel Useche





# TABLE OF CONTENTS

**01**

---

Exploratory Data  
Analysis

**02**

---

Base Models

**03**

---

Bias in the dataset

**04**

---

Debiasing Variational  
Autoencoder

**05**

---

Binary-Physical

**06**

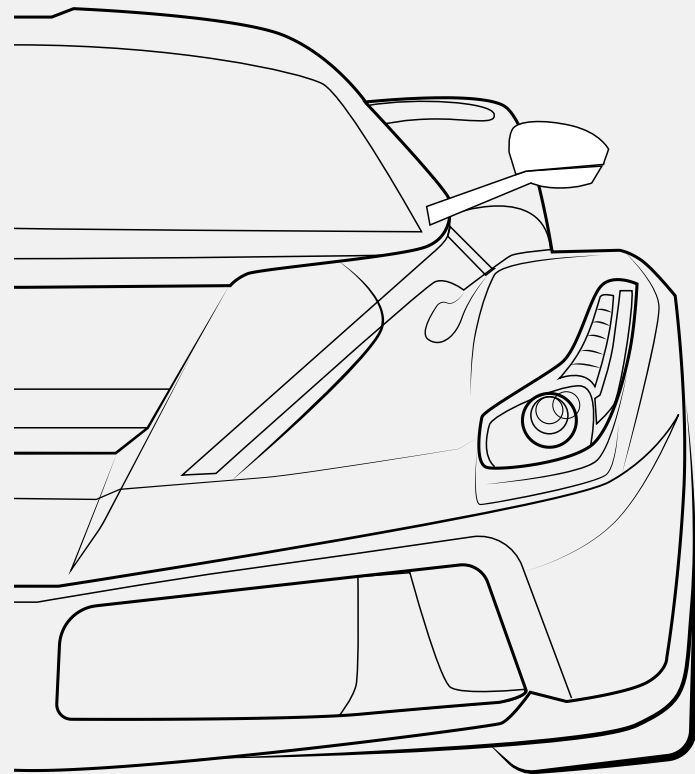
---

Conclusions

01

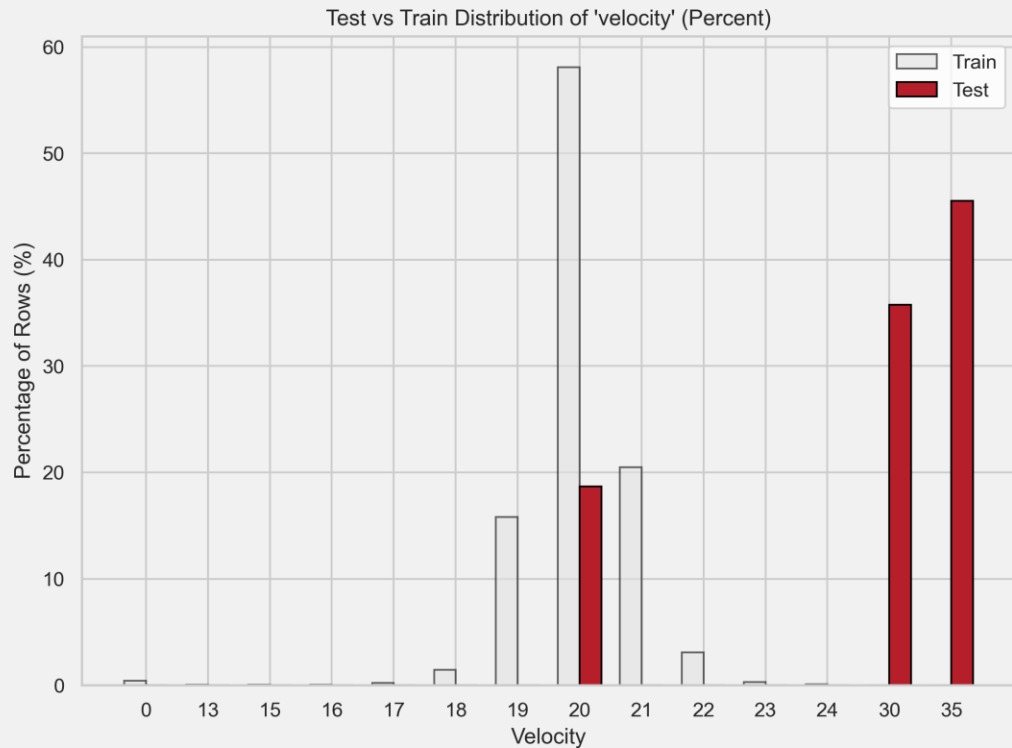
---

# EXPLORATORY **DATA ANALYSIS**



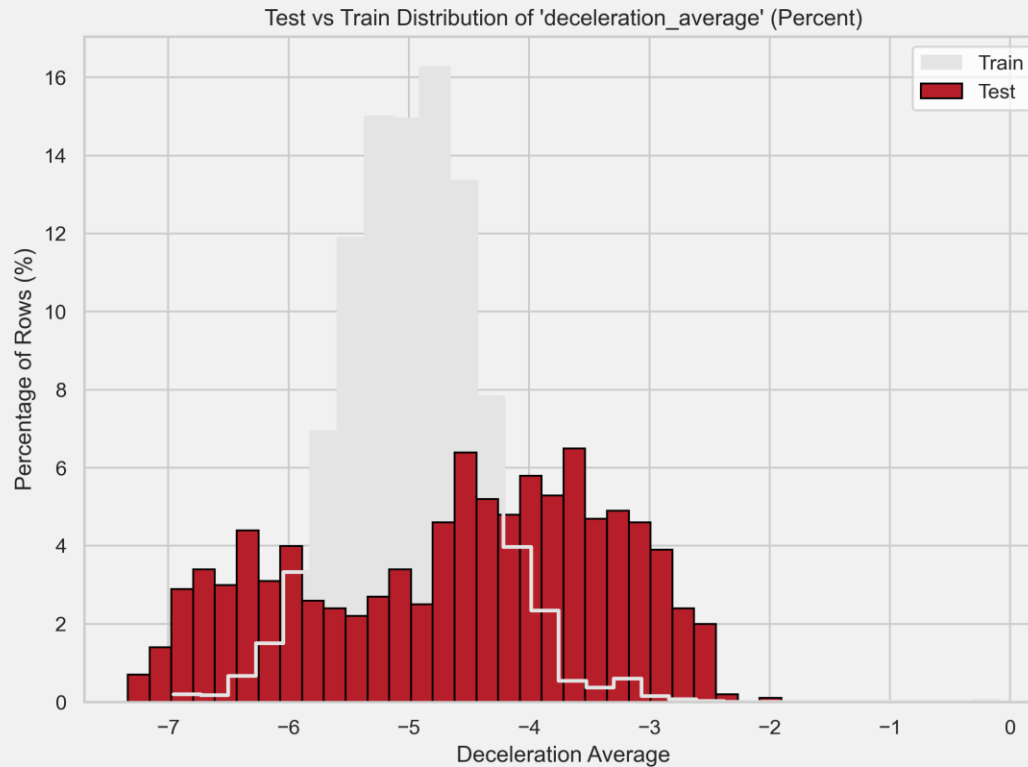


# VELOCITY **ANALYSIS**





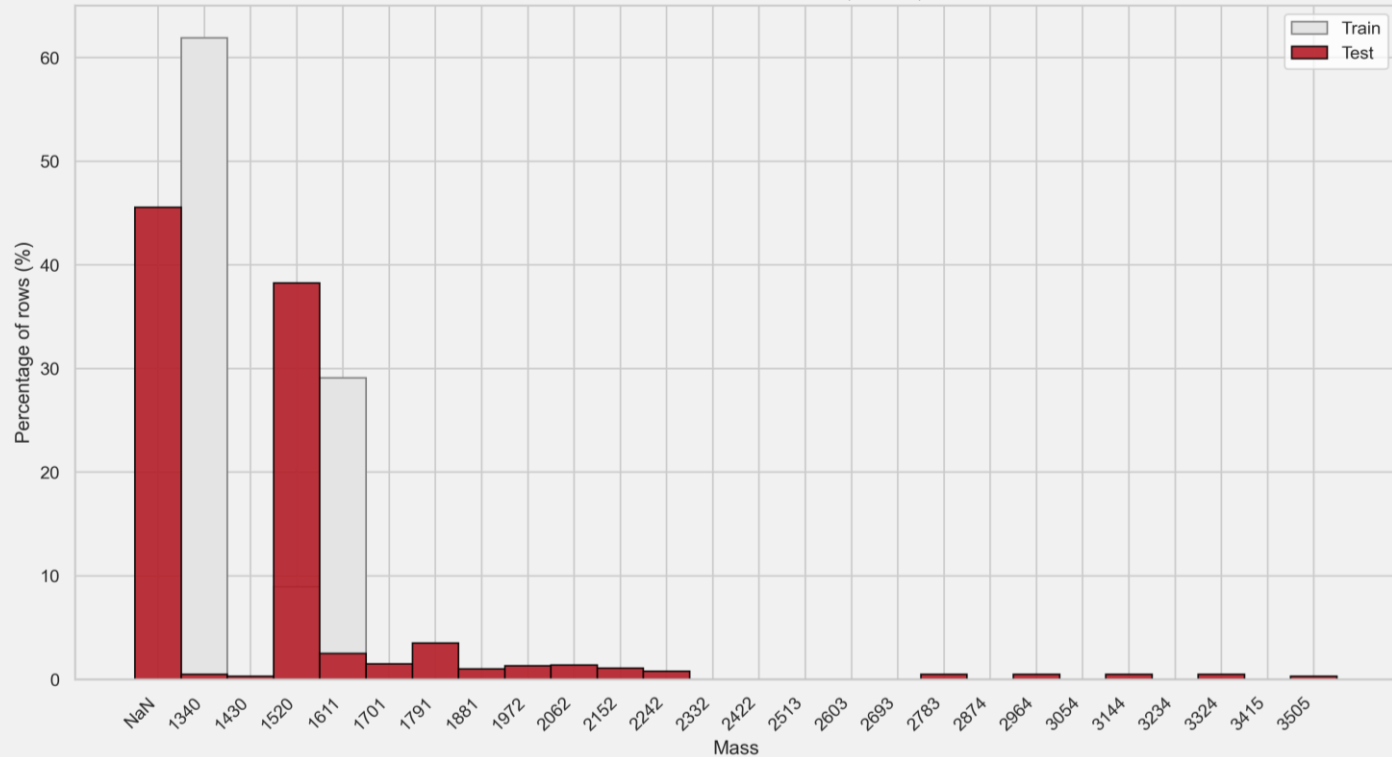
# DECELERATION AVERAGE **ANALYSIS**





# MASS ANALYSIS

Test vs Train Distribution of 'mass' (Percent)



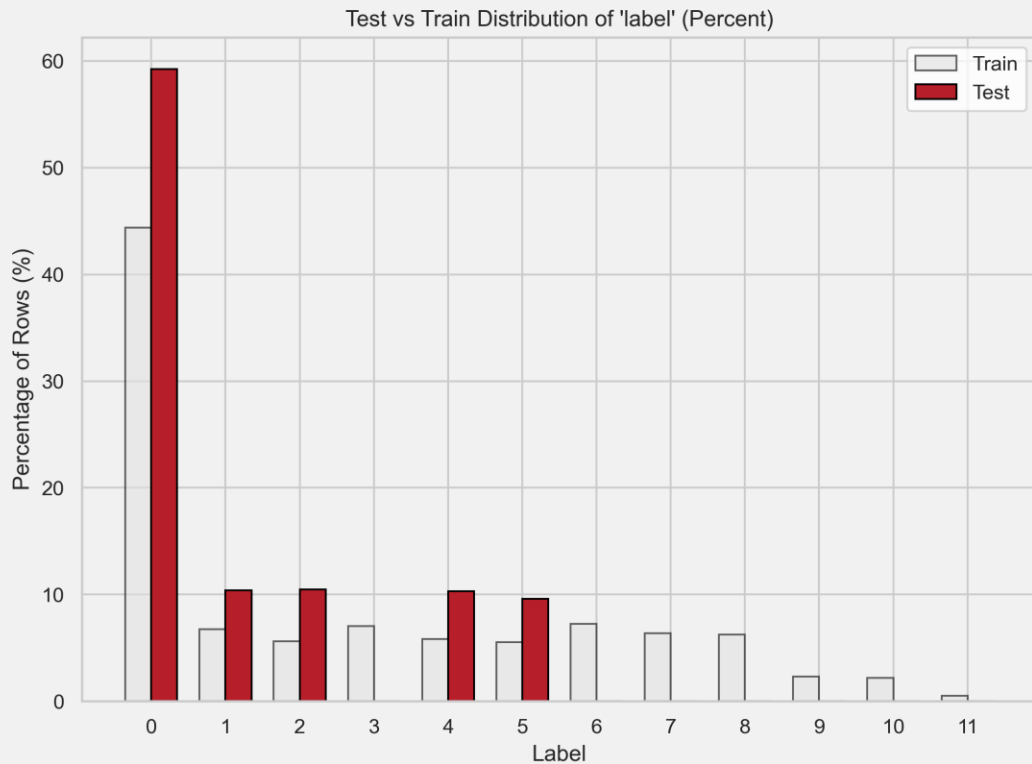
45.6%

NaN

All of them  
*fourfour*  
model



# LABEL ANALYSIS

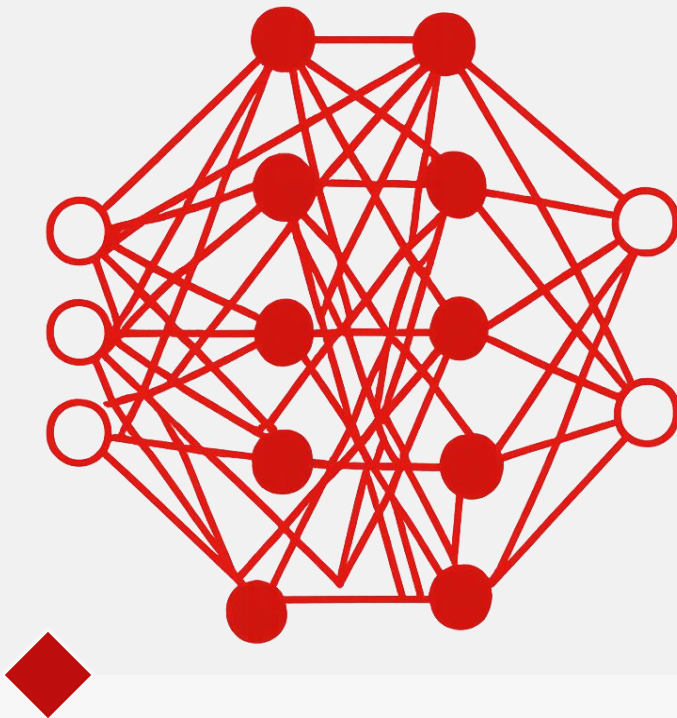


# 02

---

## BASE **MODELS**

FCN, CNN, LSTM



# BEST **MODEL**



## F1 SCORE

Harmonic mean of precision and recall,  
especially useful when dealing with  
imbalanced dataset.



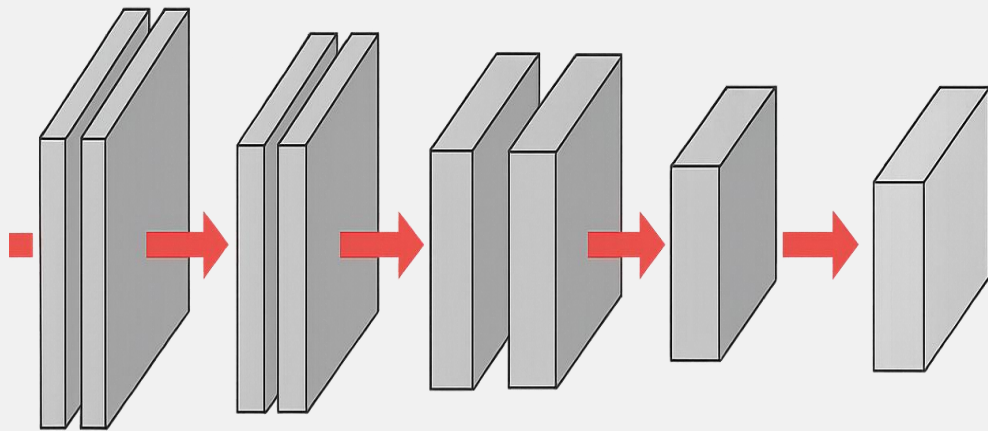
## TEST ACCURACY

Reflects how well the model generalizes to  
new, unseen data, crucial for evaluating its  
real-world usability.

# FCN

## ARCHITECTURE

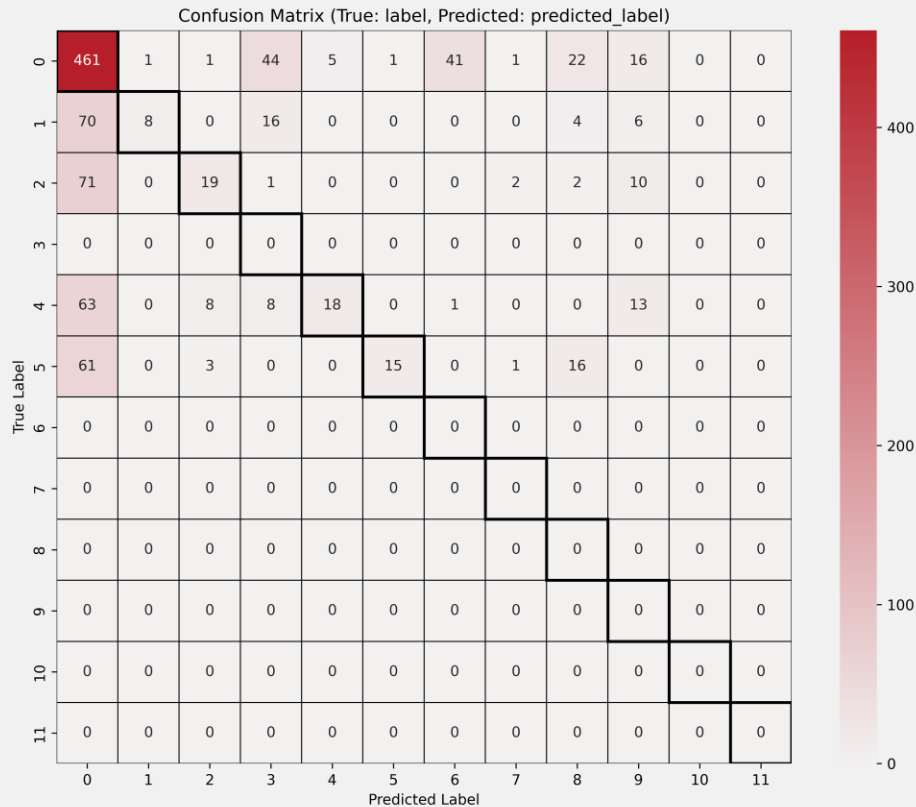
- Input layer.
- Three sequential 1D **convolutional** blocks.
- Global average **pooling** layer.
- Final fully connected (**linear**) output layer



## RESULTS

| Hyperparameter          | Value       |
|-------------------------|-------------|
| Batch size              | 32          |
| Learning rate           | $10^{-4}$   |
| Number of epochs        | 20          |
| Optimizer               | Adam        |
| Activation function     | ReLU        |
| Kernel size             | 8, 5, and 3 |
| Output channels (Conv1) | 64          |
| Output channels (Conv2) | 128         |
| Output channels (Conv3) | 128         |

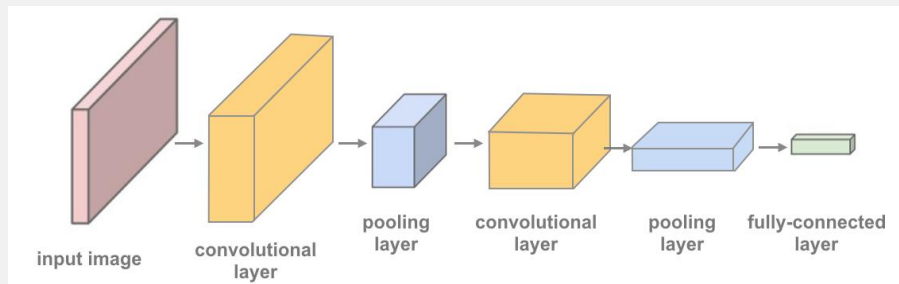
| Metrics                     | Value  |
|-----------------------------|--------|
| Average Test Accuracy       | 0.4840 |
| Average macro-F1 Score      | 0.3401 |
| Average Validation Accuracy | 0.9911 |



# 2D-CNN

## ARCHITECTURE

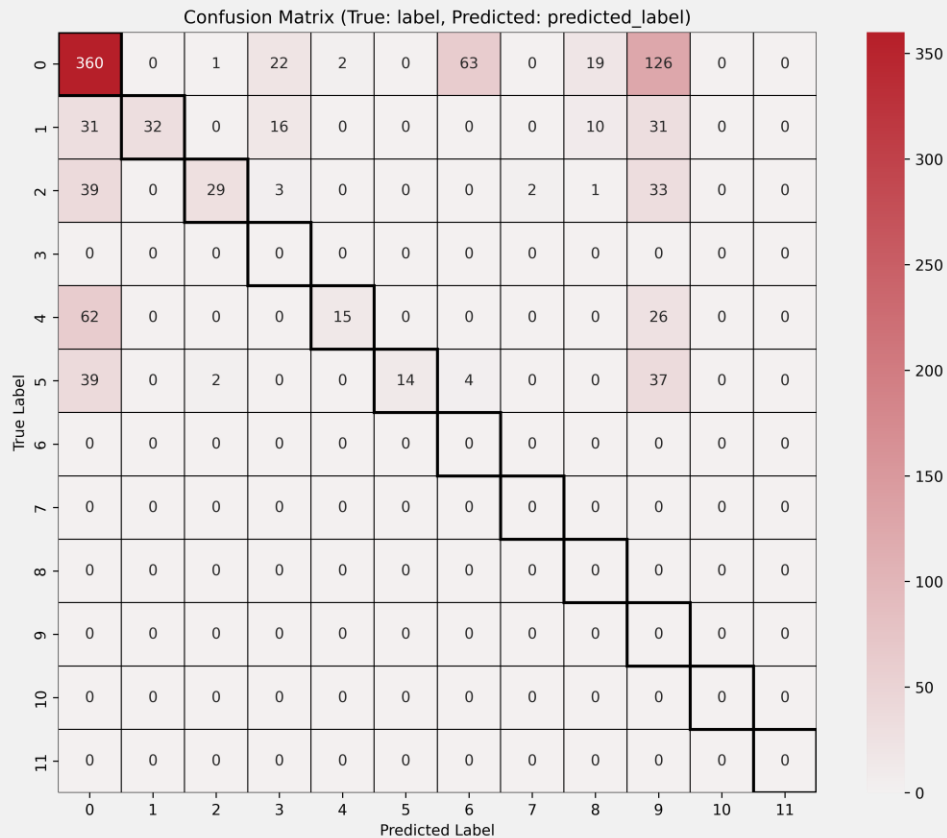
- Learn how sensors change over time **and** how they relate to each other at the same time
- **Conv2D (1×6)** → sensor correlation
- **Conv2D (3×1)** → temporal pattern
- Flatten → FC → SoftMax (12 classes)
- Conv2D followed by batch normalization, ReLU, pooling, and dropout.



## RESULTS

| Hyperparameter          | Value     |
|-------------------------|-----------|
| Batch size              | 64        |
| Learning rate           | $10^{-3}$ |
| Number of epochs        | 100       |
| Output channels (Conv1) | 32        |
| Output channels (Conv2) | 64        |
| FC hidden units         | 64        |
| Dropout rate            | 0.01      |

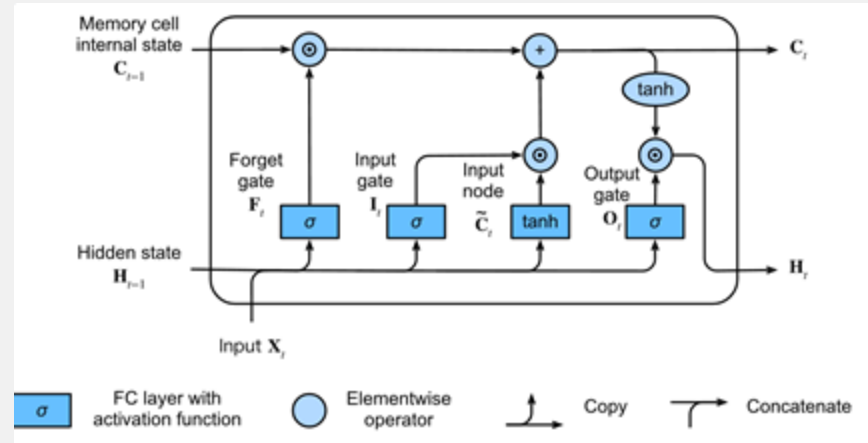
| Metrics                     | Value  |
|-----------------------------|--------|
| Average Test Accuracy       | 0.4390 |
| Average macro-F1 Score      | 0.3898 |
| Average Validation Accuracy | 0.9975 |



# LSTM

## ARCHITECTURE

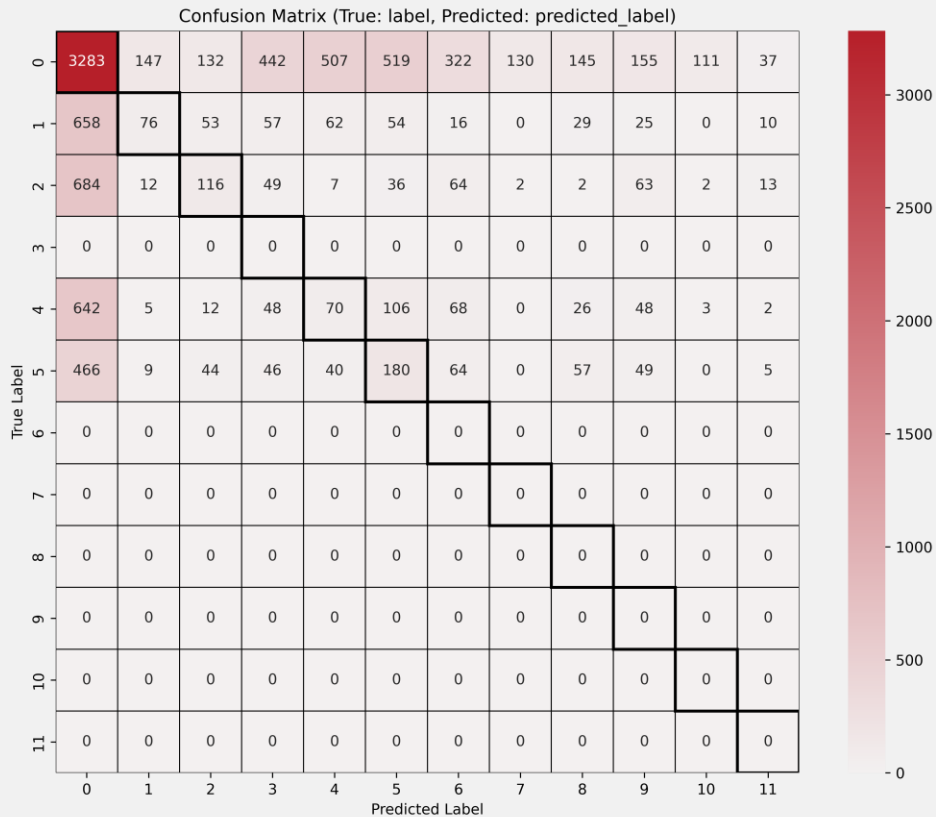
- Input: 128 timesteps  $\times$  6 sensor channels
- Gate-controlled memory (input, forget, output gates)
- Single fully-connected layer: maps summary to brake combination



## RESULTS

| Hyperparameter   | Value     |
|------------------|-----------|
| Batch size       | 64        |
| Learning rate    | $10^{-3}$ |
| Number of epochs | 40        |
| Optimizer        | Adam      |
| Number of layers | 4         |

| Metrics                     | Value  |
|-----------------------------|--------|
| Average Test Accuracy       | 0.3721 |
| Average macro-F1 Score      | 0.2165 |
| Average Validation Accuracy | 0.9973 |

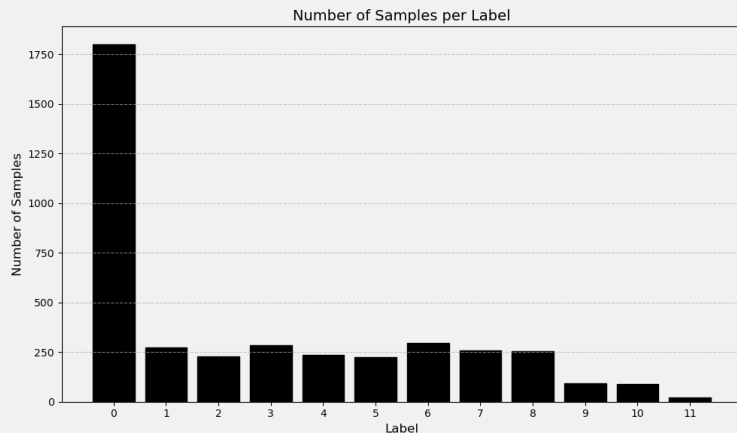


# 03

## BIAS IN THE **DATA SET**

**BIAS** reduces generalization

Class imbalance (class 0).  
Correlation with Metadata.  
Train/Test Distribution Shift.





# PREPROCESSING & DATA **AUGMENTATION**



## Standardization/Min-Max Normalization

No significant improvement in the overall performance



## Time shifting

Tests model sensitivity to minor misalignments in measurements. Slight boost test accuracy by small shifts but larger shifts impacted F1-score.



## Gaussian Noise

Adds zero-mean Gaussian noise to every sensor channel. Large noise degraded the performance.



## Scaling

Varies signal amplitude and simulates differences in sensor calibration or brake intensity. Tight scaling on az, ry, rz yielded better F1 score, whereas extreme scaling hurt accuracy.



# DATA **AUGMENTATION**



## Truncation

Removes a fixed fraction (e.g. 10-30 %) of the beginning of each signal. Even small truncation eroded performance



## Sliding window

Splits each signal into smaller windows, treats each window as an independent training example. Drastic reduction in Macro F1-score.



## Selective augmentation

Selected minority classes were over-sampled, moderate noise was introduced, down-sampling of majority class. Limited improvement in misclassification.

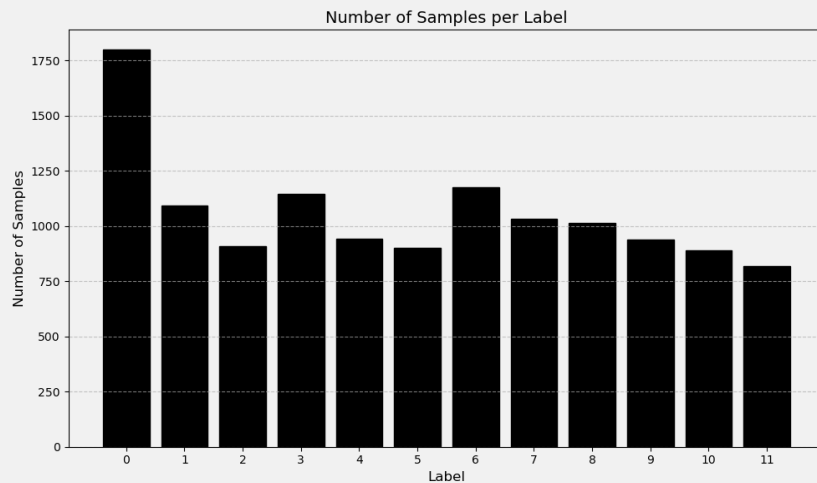
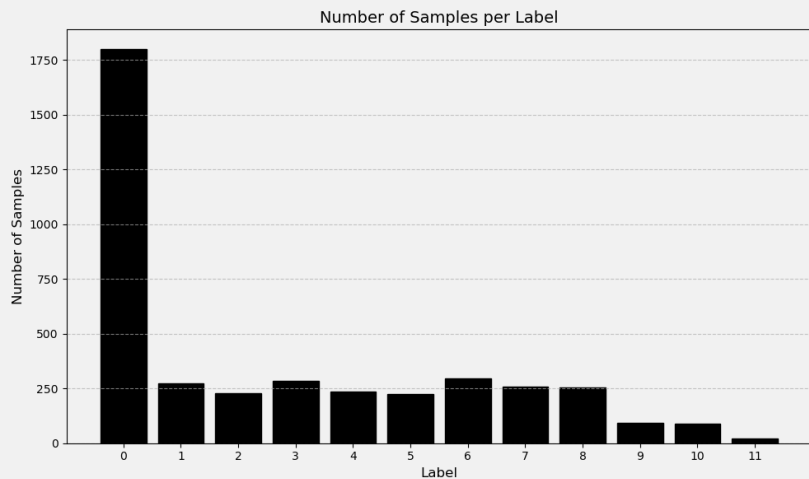


# DATA **AUGMENTATION** - RESULTS

| Augmentation                                                    | Augmentation parameter | Test accuracy | Macro F1 score |
|-----------------------------------------------------------------|------------------------|---------------|----------------|
| Gaussian Noise                                                  | 0.001                  | 0.4693        | 0.3134         |
|                                                                 | 0.005                  | 0.4739        | <b>0.3139</b>  |
|                                                                 | 0.01                   | <b>0.4939</b> | 0.3067         |
|                                                                 | 0.02                   | 0.4627        | 0.3128         |
| Time shifting                                                   | 2                      | <b>0.4702</b> | <b>0.3149</b>  |
|                                                                 | 5                      | 0.477         | 0.2973         |
|                                                                 | 10                     | 0.4654        | 0.3015         |
| Scaling (axes 3,4)                                              | (0.8,1.2)              | <b>0.4757</b> | 0.3231         |
| Scaling (axes 3,4)                                              | (0.5,2.0)              | 0.4578        | 0.2944         |
| Scaling (axes 2,4,5)                                            | (0.5,2.0)              | 0.4563        | <b>0.3256</b>  |
| Truncation                                                      | 10%                    | 0.4601        | 0.2959         |
|                                                                 | 20%                    | 0.4205        | 0.2546         |
| Sliding window (Window size x Stride)                           | 32 x 16                | 0.4921        | 0.2496         |
|                                                                 | 64 x 32                | 0.4521        | 0.2515         |
| Selective augmentation (Minority oversampling + moderate noise) |                        | 0.4851        | 0.3084         |



# Selective augmentation with different techniques



In this experiment, an **average test accuracy** over 10 random splits of 0.4301 was obtained with an **average macro F1 score** of 0.1684.

04

---

Debiasing

**Variational  
Autoencoder(VAE)**

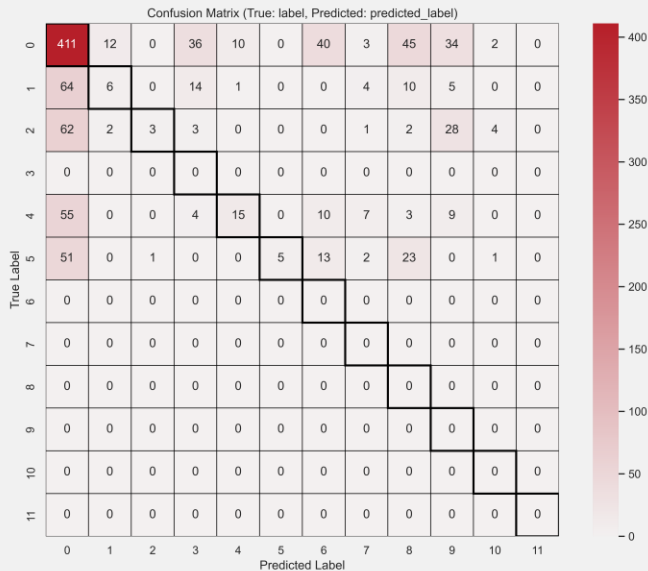


# Debiasing VAE

## ARCHITECTURE

- ♦ **FCN based**
- ♦ **Encoder:** 3 Conv1D layers followed by batch normalization and ReLU (FCN backbone)
  - Final output: latent mean ( $\mu$ ) & log-variance ( $\log\sigma^2$ )
- ♦ **Decoder:** Linear layer to reconstruct the original input ( $128 \times 6$ ) from latent vector
- ♦ **Classifier:** Fully connected layer
  - Latent vector  $\rightarrow$  12 logits
  - Train together with the encoder and decoder
- ♦ **Loss function:**  $L_{\text{total}} = \text{MSE} + \beta \cdot \text{KL} + \lambda \cdot \text{CE}$

# RESULTS



- VAE learns **slightly better**, but still biased to majority classes
- **Batch size (32), LR (0.0001) → higher** accuracy, but **lower** macro F1
- **Adaptive Sampling** (latent-based) → Focused on underrepresented regions
- **Class-weighted Loss** → Emphasized rare classes
- Both methods underperformed compared to **uniform sampling**

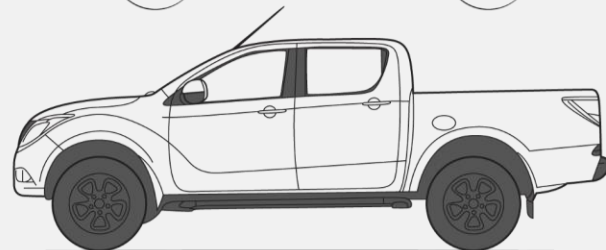
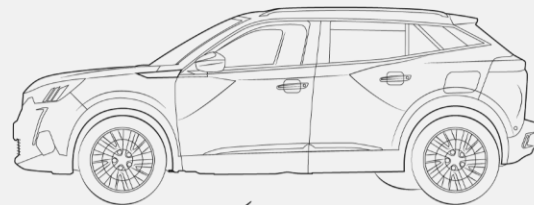
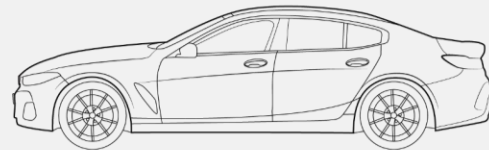
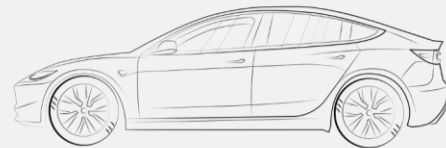
| Hyperparameter                      | Value     |
|-------------------------------------|-----------|
| Batch size                          | 64        |
| Number of epochs                    | 100       |
| Learning rate                       | $10^{-3}$ |
| Latent dimension                    | 16        |
| KL weight ( $\beta$ )               | $10^{-3}$ |
| Classification weight ( $\lambda$ ) | 1.0       |

| Model               | Test Accuracy | Macro F1      |
|---------------------|---------------|---------------|
| FCN (Baseline)      | 0.4840        | 0.3401        |
| FCN VAE             | <b>0.4861</b> | <b>0.3544</b> |
| Smaller LR & Batch  | 0.5063        | 0.2612(↓)     |
| Adaptive Sampling   | 0.4496        | 0.2947        |
| Class-weighted Loss | 0.4613        | 0.3377        |

05

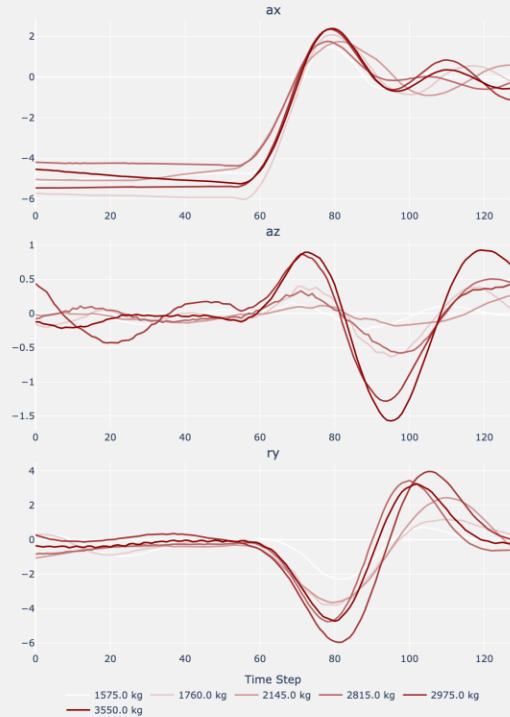
---

# BINARY - PHYSICAL **MODELS**

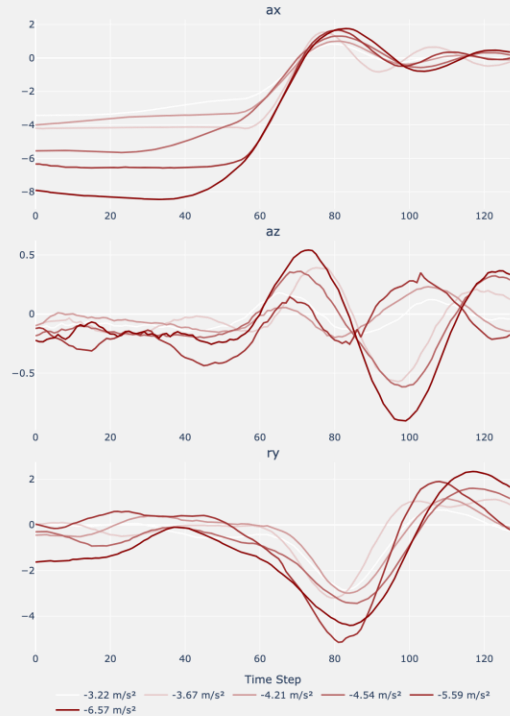


# PHYSICAL FEATURES **MODEL**

Effect of Mass on Signals



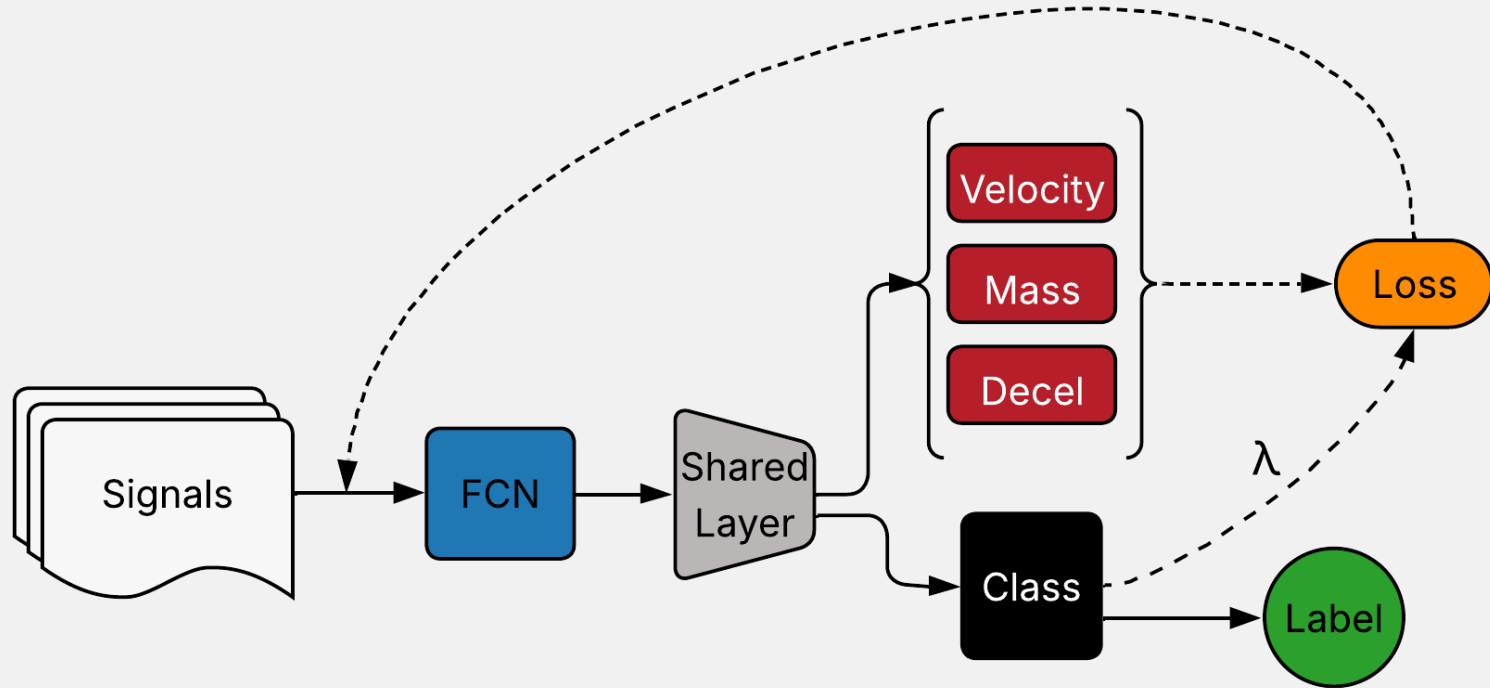
Effect of Deceleration Average on Signals



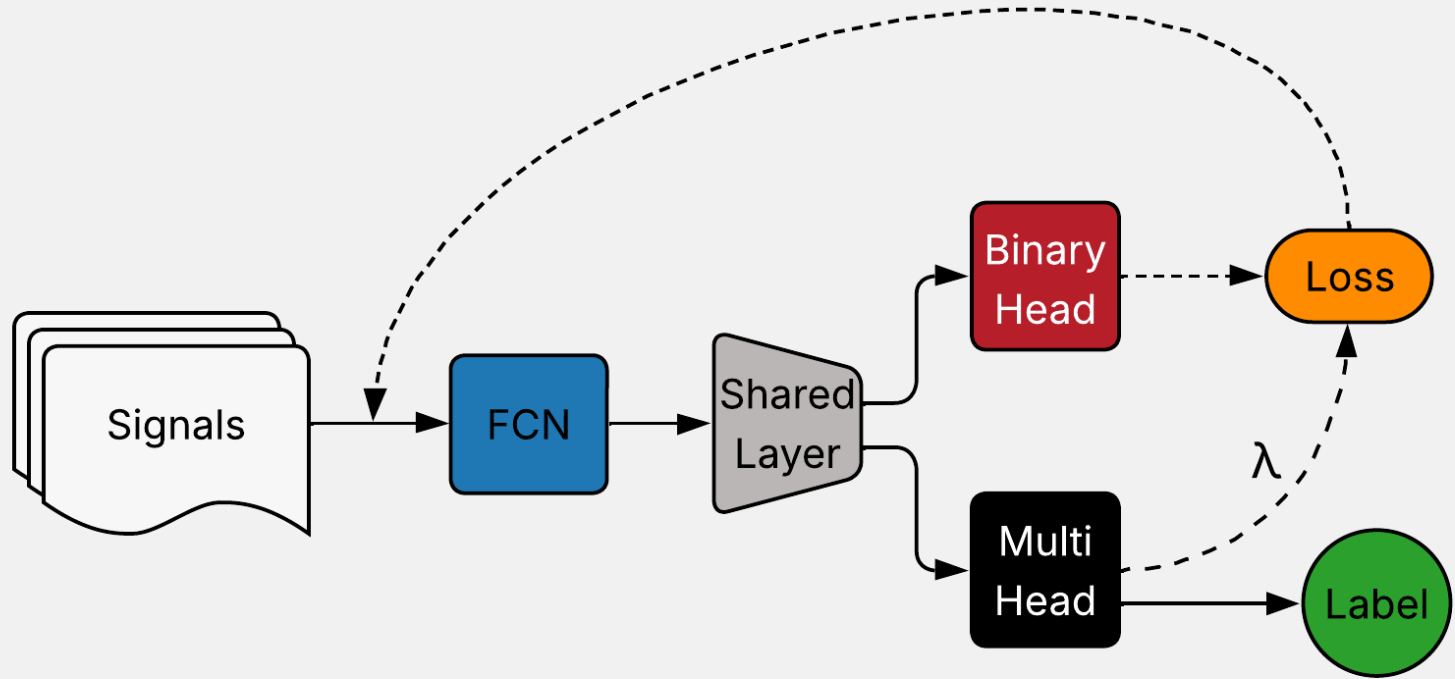
Effect of Velocity on Signals

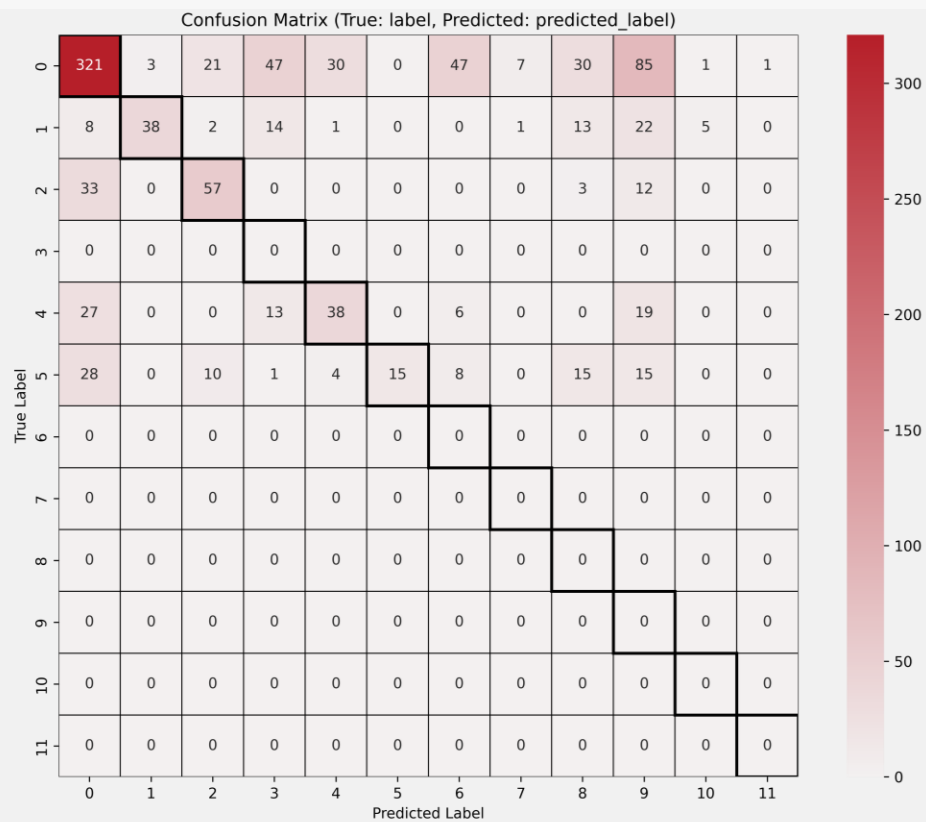


# PHYSICAL FEATURES **MODEL**



# BINARY **MODEL**





| Hyperparameter          | Value      |
|-------------------------|------------|
| Learning rate           | $10^{-3}$  |
| Number of epochs        | 5          |
| $\lambda$ (loss weight) | 1          |
| Oversampling target     | labels = 0 |
| Oversample multiplier   | 1          |
| Standardize             | True       |
| Scaling range           | (0.8, 1.8) |
| Scaled channels         | ax, az, ry |

| Metrics                     | Value  |
|-----------------------------|--------|
| Iterations                  | 20     |
| Average Test Accuracy       | 0.4319 |
| Average Test macro-F1 Score | 0.4047 |

# CONCLUSIONS



CLASS IMBALANCE



LACK OF  
GENERALIZATION



MODEST GAINS FROM:  
Augmentation  
VAE  
Additional architectures



# FUTURE **WORK**



---

METADATA INCLUSION



---

MORE DIVERSE  
TRAIN DATASET



---

MORE SOPHISTICATED  
MODELS



A person wearing a dark suit and a watch is driving a car, with their hand on the steering wheel. The background is slightly blurred, showing the interior of the vehicle. A white rectangular box is overlaid on the image, containing the text 'THANK YOU!'. A small red diamond is in the top right corner of the white box.

THANK  
**YOU!**

---